Nexa TEAM ECOBYTE

SDC HACKATHON 2026 · HOSTED BY XMUM AI CLUB

Technical Tools Declaration

Every API, cloud service, AI tool, dataset, model and sponsor resource used to build Nexa. No API keys or secrets appear in this document.

TEAM
EcoByte

PROJECT
Nexa

DOMAIN
Domain 1 · Aiversecity Future-Ready Workforce

SDG TRACK
SDG 4 · Target 4.4 (+5, 8, 9, 10)

LIVE PROTOTYPE
https://nexa-xxx.vercel.app/

FILE
SDC2026_EcoByte_TechDeclaration.pdf

1. Team and Project Overview

Team Name	EcoByte
Project / Solution Name	Nexa
Name (Leader)	Eya Hia
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University Affiliation(s)	Eya Hia — Lincoln University College Ng Wei Meng — Universiti Kebangsaan Malaysia (UKM) Alya Maisarah binti Jaafad Sadiq — Universiti Teknologi Malaysia (UTM) Mohamad Khairul Haiqal bin Mohamad Razali — Universiti Malaysia Kelantan (UMK)
Live Prototype URL	https://nexa-xxx.vercel.app/
Code Repository URL (public / access granted to judges)	https://github.com/eyahia1712/Nexa (public at submission)

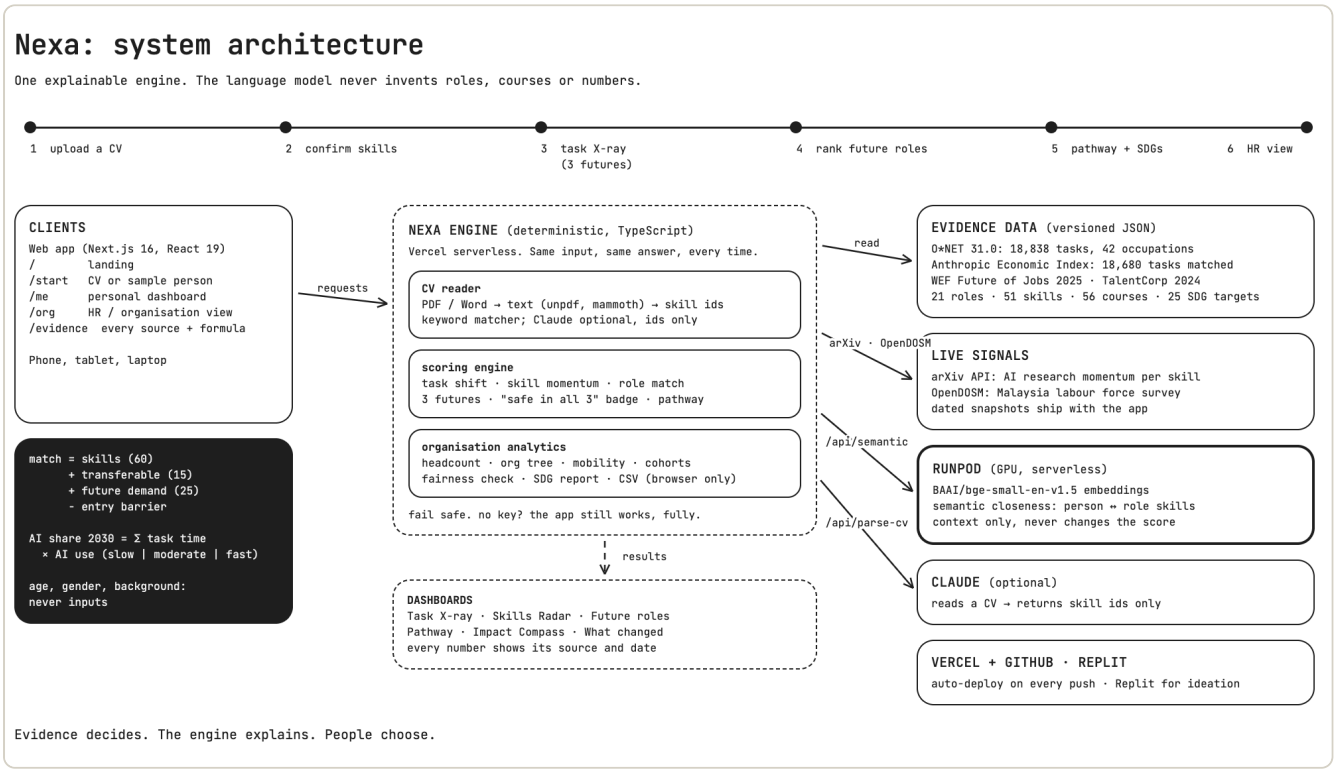
2. Architecture & Technology Stack

2.1 System Architecture Overview

> *How the components connect end to end. [Maximum 200 words]*

A person or HR user opens the Next.js web app hosted on Vercel. The browser holds the profile; nothing personal is stored on a server. A CV is sent once to `/api/parse-cv`, which extracts the text (unpdf, mammoth) and maps it to catalogue skill ids with a keyword matcher, or optionally with the Claude API. The deterministic TypeScript engine then reads versioned JSON datasets built from O*NET 31.0, the Anthropic Economic Index, WEF 2025, TalentCorp and our curated catalogue (21 roles, 51 skills, 56 courses, 25 SDG targets) to compute task shift in three scenarios, skill momentum, role matches, pathways and organisation analytics. Live signals come from the arXiv API and OpenDOSM through `/api/arxiv` and `/api/labour`, with dated snapshots as a fallback. `/api/semantic` calls a RunPod Serverless GPU endpoint running BAAI/bge-small-en-v1.5 to add semantic skill closeness beside each role. GitHub holds the code and every push auto-deploys to Vercel. There is no database: datasets ship with the build, and HR CSV files are parsed in the browser only (Appendix A).

172 / 200 words



Appendix A. System block diagram.

2. Architecture & Technology Stack

2.2 Languages, Frameworks & Libraries

Name & Version	Purpose in Project	License
TypeScript 6	Main language for app and engine	Apache-2.0
Node.js 20+	Server runtime for API routes	MIT
Python 3.11	Data build scripts, RunPod handler	PSF
Next.js 16 (App Router)	Web framework, API routes	MIT
React 19	User interface	MIT
Tailwind CSS 4	Styling	MIT
Motion 13	Animations	MIT
Zod 4	Input validation	MIT
unpdf 1.8	Read text from PDF CVs	MIT
mammoth 1.12	Read text from Word CVs	BSD-2-Clause
clsx, tailwind-merge	CSS class helpers	MIT
@anthropic-ai/sdk 0.127	Optional Claude API client (CV reading)	MIT
runpod (Python SDK) 1.7+	Serverless GPU worker	MIT
sentence-transformers 3+	Embedding model runtime on RunPod	Apache-2.0
PyTorch 2.3+	GPU tensor runtime on RunPod	BSD-3-Clause
Vitest 5, ESLint 9, eslint-config-next	Tests and linting	MIT
tsx 4, @tailwindcss/postcss 4, server-only	Script runner, CSS build, server-only guard	MIT
Google Fonts: Archivo, Archivo Narrow, Geist Mono (via next/font)	Typography	SIL Open Font License 1.1
Python standard library only (csv, json, urllib)	Data build, demo-org build and link-check scripts	PSF

3. APIs & Cloud Services

3.1 External APIs

> *Every external API the solution calls. No API keys or secrets are included.*

API and Provider	Purpose in Project	Access Tier
arXiv API (Cornell University)	AI research volume per skill, live and in dated snapshots	Free, public
OpenDOSM API (Department of Statistics Malaysia)	Malaysia labour force survey, live and in dated snapshots	Free, public
RunPod Serverless API	Semantic skill closeness on a GPU (BAAI/bge-small-en-v1.5)	Sponsor-provided (RunPod credit)
Anthropic Claude API	Optional: map CV text to catalogue skill ids	Paid; optional, off unless a key is set

3.2 Cloud Platforms & Deployment

Service & Provider	Purpose in Project	Plan / Region
Vercel	Hosting of the live prototype; serverless API routes; auto-deploy from GitHub	Hobby (free) · global edge network
RunPod Serverless	GPU endpoint for the embedding model, scales to zero	Sponsor credit (US$25) · serverless CUDA GPU, scale-to-zero
GitHub	Source code, version history, Vercel trigger	Free
Replit	Ideation with Replit Agent, pitch-deck drafts; .replit run config	Replit Core (sponsor-provided)

How the prototype is deployed: the GitHub repository is connected to Vercel; every push to `main` builds and deploys the Next.js app to https://nexa-xxx.vercel.app/. The RunPod worker is built from `runpod/Dockerfile` and deployed as a Serverless endpoint; its ID and key are stored as Vercel environment variables.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

Model & Version	Source / Provider	Usage	License / Terms
BAAI/bge-small-en-v1.5	Beijing Academy of AI, via Hugging Face	Pre-trained (inference only, on RunPod)	MIT
Claude (claude-opus-5-5)	Anthropic API	Pre-trained (optional, CV → skill ids)	Anthropic Commercial Terms
Nexa scoring engine	Team EcoByte	Rule-based, deterministic (no training)	Apache-2.0 (this repo)

4. AI Tools, Models & Datasets

4.2 Datasets

> *State clearly if a dataset contains personal or sensitive data.*

Dataset Name	Source / URL	Type	License	Personal Data? (Y/N)
O*NET 31.0 Database	onetcenter.org/database.html	Public	CC BY 4.0	N
Anthropic Economic Index (2025-03-27, 2026-03-05)	huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY	N
WEF Future of Jobs Report 2025	weforum.org	Public report (figures cited)	WEF terms, attributed	N
TalentCorp Impact Study on AI, Digital & Green Economy (2024)	talentcorp.com.my	Public report (roles cited)	Attributed	N
arXiv metadata	export.arxiv.org/api	Public API	CC0 (metadata)	N
OpenDOSM labour force survey	api.data.gov.my	Public API	CC BY 4.0	N
UN SDG Goals and targets	sdgs.un.org	Public	UN terms, attributed	N
Course catalogue (56 courses)	Public course pages, link-checked	Self-collected	Links only	N
Sample people and 37-person demo organisation (names, careers, skills)	data/curated/ (this repo)	Synthetic	Apache-2.0	N

4. AI Tools, Models & Datasets

4.2 Datasets (continued)

Dataset Name	Source / URL	Type	License	Personal Data? (Y/N)
Photos of sample people and team members	Provided by the team (our own photos)	Self-collected	Used with the owners' consent	Y (team's own images, used with consent; no other personal data)
HRD Corp course directory and e-LATiH	hrdcorp.gov.my · e-latih.hrdcorp.gov.my	Public	Linked and attributed	N
Course providers linked in the pathway	Coursera, edX, Kaggle Learn, Microsoft Learn, Hugging Face, DeepLearning.AI, Cisco NetAcad, HubSpot Academy, Harvard CS50, AWS Skill Builder, Salesforce Trailhead, freeCodeCamp, fast.ai, Khan Academy, Elements of AI, UN CC:Learn, SDG Academy, OpenWHO	Public course pages (links only)	Each provider's terms	N
Hays Asia Salary Guide 2025 (Malaysia)	hays.com.my	Public report (statistic cited)	Attributed	N
MOSTI reskilling estimate (600,000+ workers)	As cited in the SDC 2026 Domain 1 brief	Public statistic (cited)	Attributed	N

4. AI Tools, Models & Datasets

4.3 AI-Assisted Development Tools

Did the team use any generative-AI tool to help create code, design, text or other project material?

☑ Yes ☐ No

Tool & Version	Used For	Extent of Use
Claude Code (Anthropic, Claude Opus 5.5)	Writing and refactoring code, data-build scripts, tests, documentation, diagrams, pitch-video assembly	Major
Replit Agent (Replit)	Comparing six project ideas; drafting the editable pitch deck	Moderate
macOS text-to-speech	Placeholder narration in the draft pitch video (final narration is a team member's voice)	Minor

How the team reviewed, tested and took responsibility for AI-generated output

> *[Maximum 100 words]*

Every AI-assisted change was read and approved by a team member before it was committed. The engine is covered by automated tests (Vitest) for scoring, scenarios and catalogue integrity, plus TypeScript and ESLint checks on every build. Each number in the app was traced back to its dataset, every course link is checked by script, and roles, courses and statistics come only from cited sources, never from model output. We tested every screen by hand on phone and laptop, and the team can explain each formula, which is also published on the app's /evidence page.

95 / 100 words

5. Sponsor-Provided Resources

Did the project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit Core plan and AI credits	Ideation with Replit Agent (six ideas compared), editable pitch deck drafts; repository runnable on Replit
RunPod	US$25 GPU credit; Serverless endpoint	Hosts the BAAI/bge-small-en-v1.5 embedding model for semantic skill closeness, called from /api/semantic

Was the sponsor resource essential to the core functionality, or could it be replaced?

> *[Maximum 100 words]*

Neither resource is needed for Nexa's core flow; this is deliberate, so the prototype never breaks for judges. Replit was essential to our process: we compared ideas and drafted the pitch deck with Replit Agent, and the repository runs on Replit through its `.replit` file. RunPod powers a GPU-only feature, semantic skill closeness with an embedding model, which complements the explainable score; without it, Nexa hides that one indicator and everything else still works. Both could be replaced (for example by another GPU host), but RunPod's serverless scale-to-zero keeps our cost close to zero.

94 / 100 words

Proof of use is attached as **SDC2026_EcoByte_Replit_POU.pdf** (team's own Replit account, projects and credit usage) and **SDC2026_EcoByte_Runpod_POU.pdf** (RunPod workload, API and worker code).

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

> *[Maximum 150 words]*

Nexa needs no account and has no database. A CV is sent once over HTTPS to our `/api/parse-cv` route, read in memory and discarded; only the suggested skills return to the browser. If the optional Claude API is switched on, the CV text is sent to Anthropic only to map it to skill ids, under Anthropic's commercial terms. Profiles, plans and progress stay in the user's own browser storage and can be cleared at any time. HR CSV files are parsed in the browser and never uploaded. Only skill names, with no names or contact details, are sent to the RunPod endpoint, and the live-signal routes send no user data at all. Every sample person and the 37-person demo organisation are synthetic. API keys live in Vercel environment variables, never in the repository or the browser.

136 / 150 words

6.2 Known Limitations & Risks

> *[Maximum 150 words]*

Forecasts rely on US task data (O*NET) and on AI use observed on one provider's platform (Anthropic Economic Index), so they may differ for Malaysian jobs; we show ranges and three scenarios, and label the scenario rates as assumptions. The catalogue covers 21 roles and 56 courses; anything outside it is not recommended rather than invented. Keyword CV reading can miss skills, so the user confirms every skill. The optional Claude step could mis-read text, so it may only return catalogue ids, which the user also confirms. Live APIs (arXiv, OpenDOSM) can rate-limit or fail, so dated snapshots ship as a fallback. RunPod cold starts add latency, so semantic closeness loads after the main results and is never part of the score. The fairness check flags gaps but cannot prove the absence of bias; age and gender are never scoring inputs.

141 / 150 words

7. Compliance & Readiness

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

Eya Hia

Eya Hia, Team Leader, EcoByte

[Date]

24 October 2026